Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Warehouse Traffic Control: 3D Distributed Systems Simulator | *JavaScript, Three.js, Web Workers, Vite*

- Built a live 3D browser simulator of a 200-robot warehouse in Three.js, deployed to GitHub Pages.
- Designed a lease-based protocol with **zero collisions** over 416K moves, even with lost messages, clock drift and crashes.
- Added replay that rewinds to any moment instantly, and ran benchmark simulations in parallel with Web Workers.

Negotiate, Then Solve: Full-Stack Scheduling Platform | *React, TypeScript, FastAPI, Python, SQLite*

- Built a React + TypeScript frontend over a FastAPI backend, with role-based permissions, an approval queue and audit history.
- An LLM turns scheduling requests from email, chat and the web into constraints for a solver.
- Tested with **215 pytest tests**; safety checks stopped all 50 prompt-injection attempts.

BoardBuddy: Real-Time Collaborative Whiteboard | *Next.js, TypeScript, Convex, WebSockets*

- Built a real-time collaborative whiteboard with live cursors, drawing, shapes, sticky notes and undo/redo.
- Synced edits between users over WebSockets and stored boards in Convex.
- Checked every read and write on the server, so users can only access their own organisation's boards.

Peer-to-Peer Distributed File Sharing System | *C++, POSIX Sockets, Multithreading, OpenSSL, Linux*

- Built a BitTorrent-style file sharing system in C++ with a central tracker, multithreaded peers and per-piece integrity checks.
- Profiled and fixed bottlenecks in scheduling and networking, cutting a 1GB download from **193 s to under 13 s**.
- Added a replicated backup tracker with automatic failover, tested with unit tests under sanitizers.

Technical Skills

Languages: TypeScript, JavaScript, Python, C/C++, SQL, HTML/CSS.
Frontend: React, Next.js, Tailwind CSS, Three.js.
Backend: FastAPI, Node.js, PostgreSQL, SQLite, Convex, Prisma.
Tools: Git, GitHub Actions, Docker, Vercel, Linux, pytest.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

Interests

- Reading web novels and manga.
- Playing chess and badminton.